Milestone 1: Airline Customer Holiday Booking

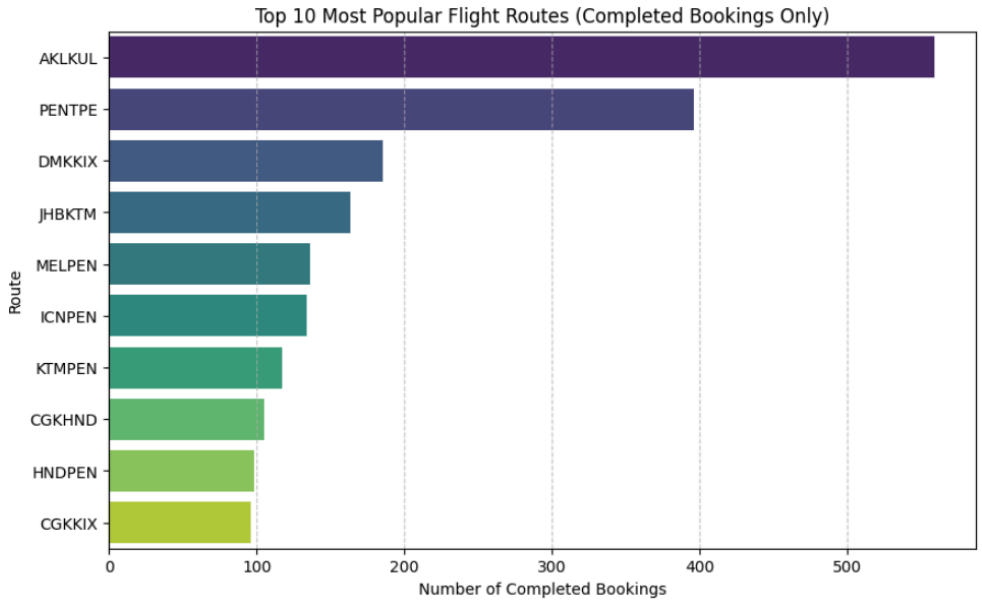
Data Engineering Questions

1. What are the top 10 most popular flight routes?

The analysis of completed bookings reveals a significant concentration of traffic on a few key routes. The data clearly shows that not all routes have equal popularity.

The most popular route by a large margin is AKLKUL, with 559 completed bookings. This is followed by PENTPE at 396 bookings. There is a notable drop-off to the third-place route, DMKKIX (185 bookings), which suggests a strong and established demand for the top two routes.

The remaining routes in the top 10 show a more gradual decline in booking volume, from JHBKTM (163) down to CGKKIX (96).



The top route, AKLKUL (Auckland to Kuala Lumpur), dominates, with over 40% more completed bookings than the second-place route. Furthermore, PEN (Penang) appears as either an origin or destination in five of the top ten routes, establishing it as a critical regional hub.

2. What is the distribution of bookings across flight hours?

The distribution of flight bookings throughout the day is not uniform, showing distinct patterns of high and low activity. The data reveals two primary peak periods for travel.

- Morning Peak (6 AM - 9 AM): Demand for flights builds rapidly in the early morning, reaching its highest point at 9 AM with 507 completed bookings.
- Afternoon Peak (12 PM - 2 PM): A secondary surge in activity occurs around midday, with booking numbers staying consistently high through 2 PM.

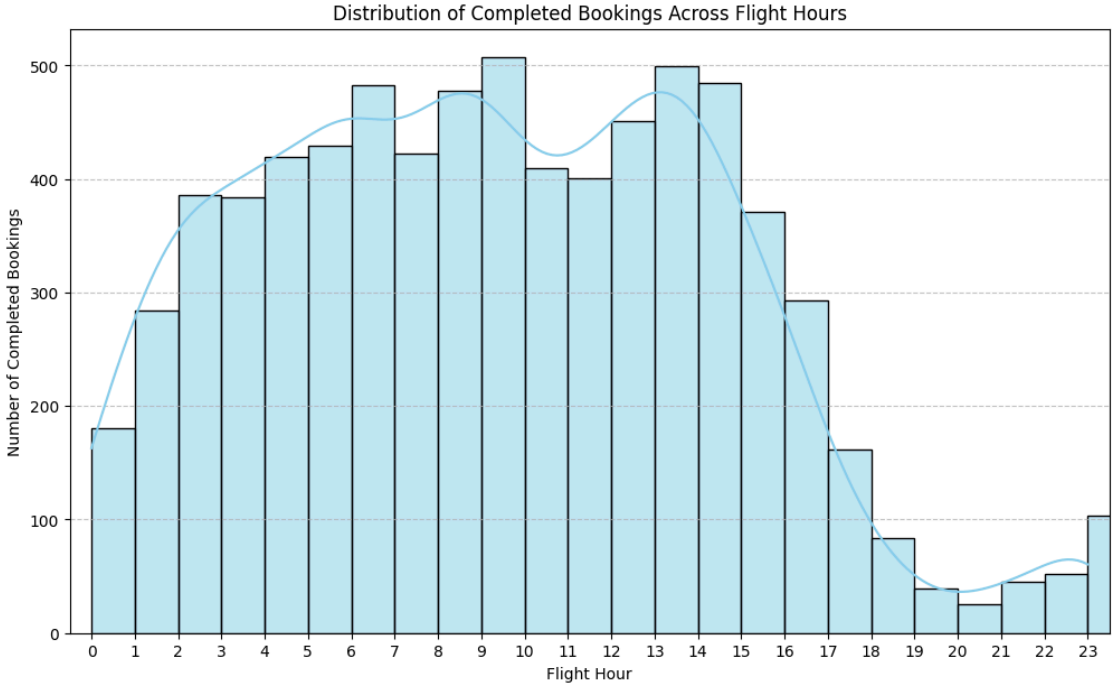
Following the afternoon peak, booking demand drops sharply after 3 PM. The lowest activity is observed in the late evening, specifically between 7 PM and 10 PM, suggesting that there are either fewer scheduled flights or a lower passenger preference for traveling at these times.The analysis of flight booking distribution throughout the day reveals a non-uniform pattern, characterized by distinct periods of high and low activity. This data points to two primary peak periods for air travel, indicating specific times when passenger demand is significantly higher.

The first major surge in activity, referred to as the Morning Peak, spans from 6 AM to 9 AM. During this three-hour window, the demand for flight bookings builds rapidly from the early morning hours, reaching its apex at 9 AM. At this specific time, a remarkable 507 completed bookings are recorded, signifying a substantial preference for morning departures. This pattern suggests that a considerable segment of travelers, possibly business travelers or those aiming to reach their destinations early, prioritize flights during this period.

Following the morning rush, a secondary but equally significant surge in activity is observed during the Afternoon Peak, occurring between 12 PM and 2 PM. During these two hours, booking numbers remain consistently high, indicating a sustained level of demand. This midday peak could be attributed to a variety of factors, including travelers who prefer to start their journeys after the morning commute, or those connecting from earlier flights.

Conversely, the data also highlights periods of significantly lower booking demand. Immediately following the afternoon peak, a sharp drop in booking activity is observed after 3 PM. This decline continues into the late evening, with the lowest activity recorded specifically between 7 PM and 10 PM. This substantial decrease in bookings during these hours suggests multiple factors, potentially including fewer scheduled flights during the late evening, or a lower passenger preference for traveling at these times due to personal or logistical reasons. The marked difference between peak and off-peak periods underscores the dynamic nature of flight booking

patterns and provides valuable insights for optimizing flight scheduling and resource allocation.



3. By analyzing review patterns, which traveler type and class combination yields the highest and lowest ratings?



The analysis of customer ratings reveals significant variations in satisfaction across different traveler types and flight classes.

- Highest Rated: The most satisfied customers are Solo Leisure travelers flying in First Class, who provide an average rating of 6.63. This segment consistently reports high satisfaction, suggesting that the airline's product and service are well-aligned with the expectations of individual travelers.
- Lowest Rated: The least satisfied group is the 'Various' traveler type in Economy Class, with an average rating of just 1.00.

Further insights from the data include:

- Business Traveler Dissatisfaction: Business travelers report notably low satisfaction regardless of class. This may indicate a failure to meet their specific needs, such as punctuality, efficiency, and onboard productivity features.
- Family Service Gap: An interesting paradox emerges where Family Leisure travelers in First Class (4.23) rate their experience lower than Solo Leisure travelers in Economy Class (4.56). This suggests that the premium service offering may not be adequately tailored to the unique needs of families traveling together.

Based on the analysis, Solo Leisure travelers in First Class reported the highest satisfaction with an average rating of 6.63. Conversely, 'Various' traveler types in Economy Class demonstrated the lowest satisfaction, with an average rating of 1.00.

Features used in the predictive model and why each feature was selected

1. **Traveller Type :** Different traveller types (e.g., business, leisure, solo, family) have distinct needs and tolerances. A business traveller may prioritize on-time performance and efficient service, while a family may value comfort, amenities, and baggage policies. The satisfaction metric changes based on meeting these specific needs.

2. **Class :** This is a primary driver of satisfaction, directly dictating the quality of the product consumed. Higher classes (First, Business) expect superior space, food, and dedicated service; failure to deliver on these premium

expectations leads to a large drop in satisfaction. Economy class customers primarily seek value, basic comfort, and timely arrival.

3. **Verified :** While not directly affecting the experience, a Verified flag confirms the feedback's authenticity. This allows the airline to trust the sentiment score and prioritize feedback from real customers over potential noise or unverified sources. This leads to more focused and effective service improvements, which ultimately drives satisfaction.

4. **Route :** affects satisfaction due to variations in flight duration, weather, airport facilities, time zones, and layovers.Long-haul routes might lower satisfaction due to fatigue or delays.Certain routes may have better service or newer aircraft, increasing satisfaction.Regional or domestic routes may have different standards or expectations altogether.

5. **Sentiment score :** This is the *outcome* being measured but also acts as a predictor when used to train a model to understand *drivers* of satisfaction. The score is a quantifiable measure of emotional state related to the flight. It directly indicates satisfaction, allowing the model to link positive/negative sentiment with the other features and identify which operational variables need improvement.

Plotting and explaining Shap and Lime analysis

While our neural network can make accurate predictions, its internal logic is complex and not inherently understandable. This is often called the "black box" problem. To build trust and gain actionable insights, we use explainability techniques like LIME and SHAP to understand *why* the model makes a specific prediction.

LIME (Local Interpretable Model-agnostic Explanations)

LIME answers the question: "Why did the model make *this specific* prediction?"

It works by creating a simple, interpretable model (like a basic linear model) that approximates the behavior of our complex neural network in the local "neighborhood" of a single data point. It's like asking a brilliant but complicated expert (our model) to explain their reasoning for one case in very simple terms.



This plot explains a single prediction where the model decided the customer was Satisfied (with a 52% probability).

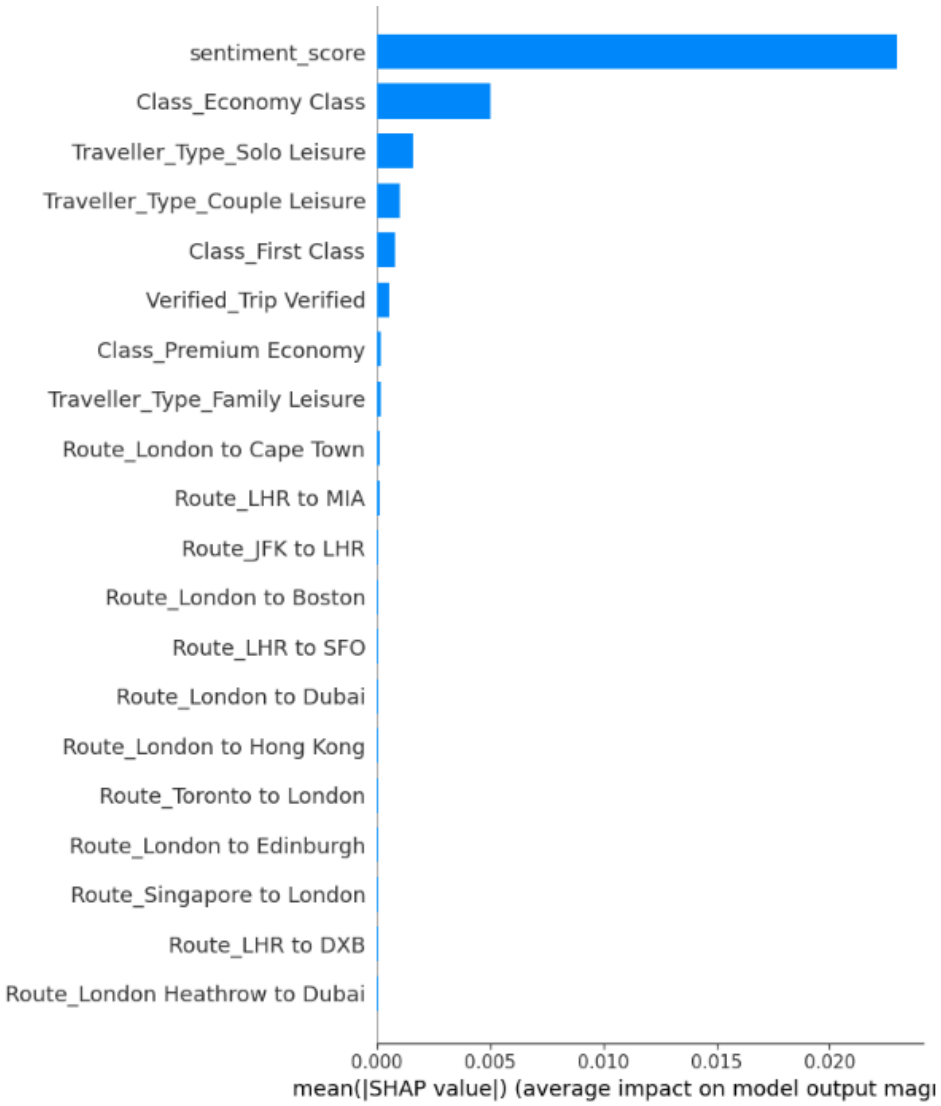
- Prediction Probabilities: The model was not very confident, predicting a 52% chance of "Satisfied" versus a 48% chance of "Not Satisfied."
- Feature Contributions: The middle section shows which features contributed to this prediction. Features in orange pushed the prediction towards "Satisfied," while features in blue would have pushed it towards "Not Satisfied."
- Interpretation:
    - The most influential factor was `sentiment_score > 0.66`. Since this customer's `sentiment_score` was 0.83 (as seen on the right), this feature strongly pushed the prediction towards "Satisfied."
    - The `Traveller_Type_Couple Leisure = 1` and `Class_Economy Class = 1` also had a positive (orange) but much smaller impact.
    - This explanation tells us that for this specific customer, the high sentiment score was the deciding factor that tipped the scales towards a "Satisfied" prediction, overcoming the other, less impactful features.

SHAP (SHapley Additive exPlanations)

SHAP is a more advanced technique rooted in game theory. It explains every prediction by quantifying how much each feature contributes to pushing the model's output away from a base value (the average prediction across the entire dataset). SHAP provides both rich *global* explanations (how the model behaves overall) and precise *local* explanations (for a single prediction).

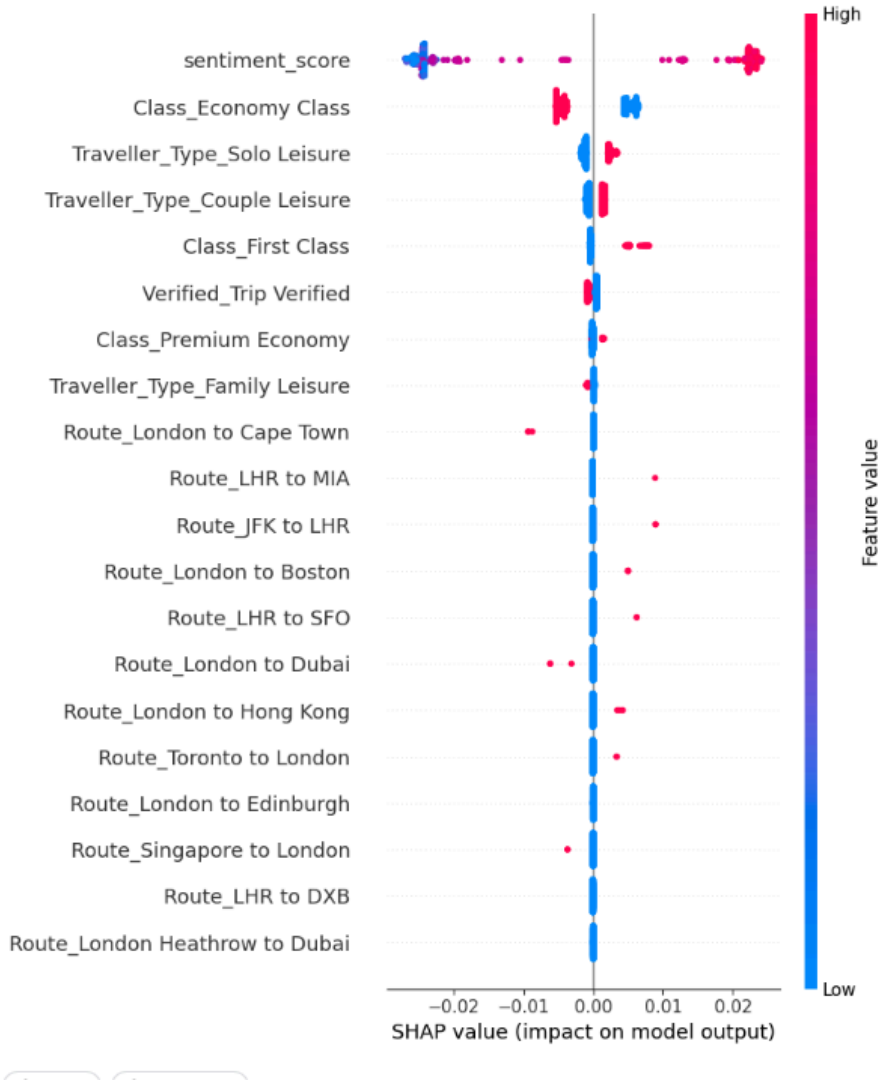
Global Explanations: Understanding Overall Model Behavior

These plots summarize the model's behavior across all data points.



This plot ranks the features by their average impact on the model's predictions, acting as a more robust feature importance chart.

- Interpretation: It clearly shows that `sentiment_score` is, on average, the most influential feature by a large margin. `Class_Economy Class` is the second most important, followed by `Traveller_Type_Solo Leisure`. This tells us which features the model relies on most heavily across the board.



This plot is much richer. It shows not only the magnitude of a feature's impact (its position on the x-axis) but also the *direction* of that impact based on the feature's value (color). Each dot represents one prediction.

- Interpretation:
  - `sentiment_score`: There is a clear trend. High sentiment scores (red dots) have positive SHAP values, pushing the prediction towards "Satisfied." Low scores (blue dots) have negative SHAP values, pushing it towards "Not Satisfied."
  - `Class_Economy Class`: The red dots (where `Class_Economy Class = 1`, meaning the passenger *is* in Economy) are clustered on the negative side. This means flying Economy consistently pushes the model's prediction towards "Not Satisfied."
  - `Traveller_Type_Solo Leisure`: The red dots (where `Traveller_Type_Solo Leisure = 1`) are mostly on the positive side, indicating that being a solo leisure traveler tends to push the prediction towards "Satisfied."

Local Explanation: Deconstructing a Single Prediction



This force plot explains a single prediction in detail. It shows how different features "push" and "pull" the prediction from the `base value` to its final score.

- Base Value (0.4693): This is the average prediction if we knew nothing about the passenger.
- Final Output `f(x)` (0.47): This is the model's final prediction for this specific passenger. It's very close to the base value.
- Forces:
    - Pushing Higher (Red): The feature `Traveller_Type_Solo Leisure = 1` is a positive force, pushing the prediction towards "Satisfied."
    - Pushing Lower (Blue): The features `sentiment_score`, `Class_Economy Class`, and `Verified_Trip` are all negative forces, pushing the prediction towards "Not Satisfied." The size of each block shows the magnitude of its impact.
- Conclusion: For this specific traveler, the positive impact of being a "Solo Leisure" type was almost perfectly canceled out by the negative impacts of their sentiment score and economy class ticket, resulting in a final prediction that is nearly neutral.